

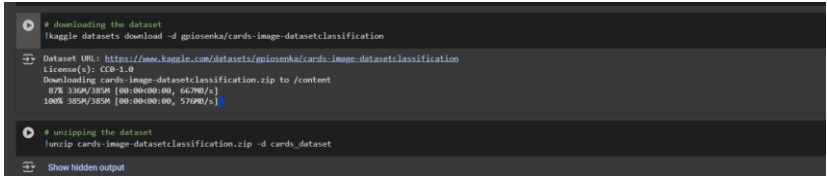
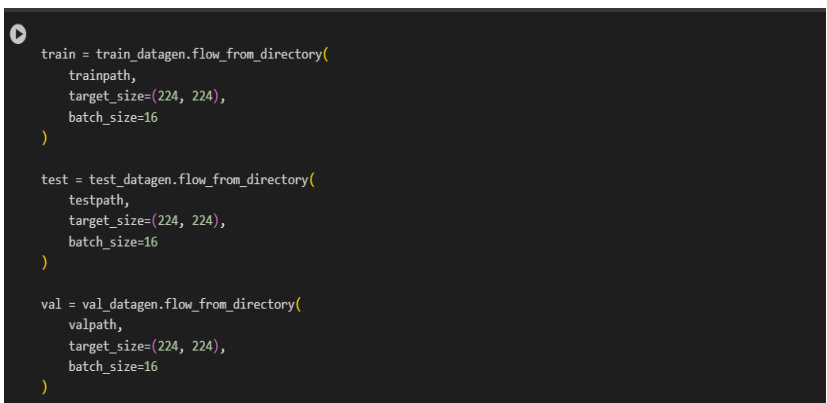
Data Collection and Preprocessing Phase

Date	19 June 2025
Team ID	SWTID1749908722
Project Title	CardMaster: Intelligent Playing Card Recognition using Transfer Learning
Maximum Marks	6 Marks

Preprocessing Template

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Section	Description
Data Overview	The dataset used consists of images of playing cards (from the Playing Cards Dataset). Each image represents a unique card in different backgrounds. These are used to train a model for card recognition.
Resizing	All images are resized to a uniform size of (224, 224) to ensure consistency and compatibility with the neural network input requirements.
Normalization	Pixel values are scaled between 0 and 1 using $\text{img} = \text{img} / 255.0$. This helps in faster and stable training.
Data Augmentation	ImageDataGenerator is used to apply augmentation like rescaling, zooming, shearing, and horizontal flipping to improve model generalization.

Data Preprocessing Code Screenshots	
Loading Data	 <pre># downloading the dataset !kaggle datasets download -d gpiosenka/cards-image-datasetclassification Dataset URL: https://www.kaggle.com/datasets/gpiosenka/cards-image-datasetclassification License(s): CC0-1.0 Downloading cards-image-datasetclassification.zip to /content 87% 336M/385M [00:00<00:00, 667MB/s] 100% 385M/385M [00:00<00:00, 576MB/s] # unzipping the dataset !unzip cards-image-datasetclassification.zip -d cards_dataset Show hidden output</pre>
Resizing	 <pre>train = train_datagen.flow_from_directory(trainpath, target_size=(224, 224), batch_size=16) test = test_datagen.flow_from_directory(testpath, target_size=(224, 224), batch_size=16) val = val_datagen.flow_from_directory(valpath, target_size=(224, 224), batch_size=16)</pre>
Normalization	 <pre>rescale = 1/255, # normalising the image</pre>

Data Augmentation

```
# Data Augmentation:
# import tensorflow as tf
# data_augment = tf.keras.Sequential([
#     layers.RandomFlip("vertical"),
#     layers.RandomRotation(0.1),
#     layers.RandomZoom(0.1),
#     layers.RandomContrast(0.1)
# ])

# Splitting the Dataset :
trainpath = "cards_dataset/train"
testpath = "cards_dataset/test"
valpath = "cards_dataset/valid"

# Data Augmentation using imageDataGenerator :
train_datagen = ImageDataGenerator(
    rescale = 1/255, # normalising the image
    zoom_range = 0.2, # zoom
    shear_range = 0.2, # stretch and contract
    vertical_flip=True #flip vertically
)
test_datagen = ImageDataGenerator(rescale=1/255)
val_datagen = ImageDataGenerator(rescale=1/255)
```